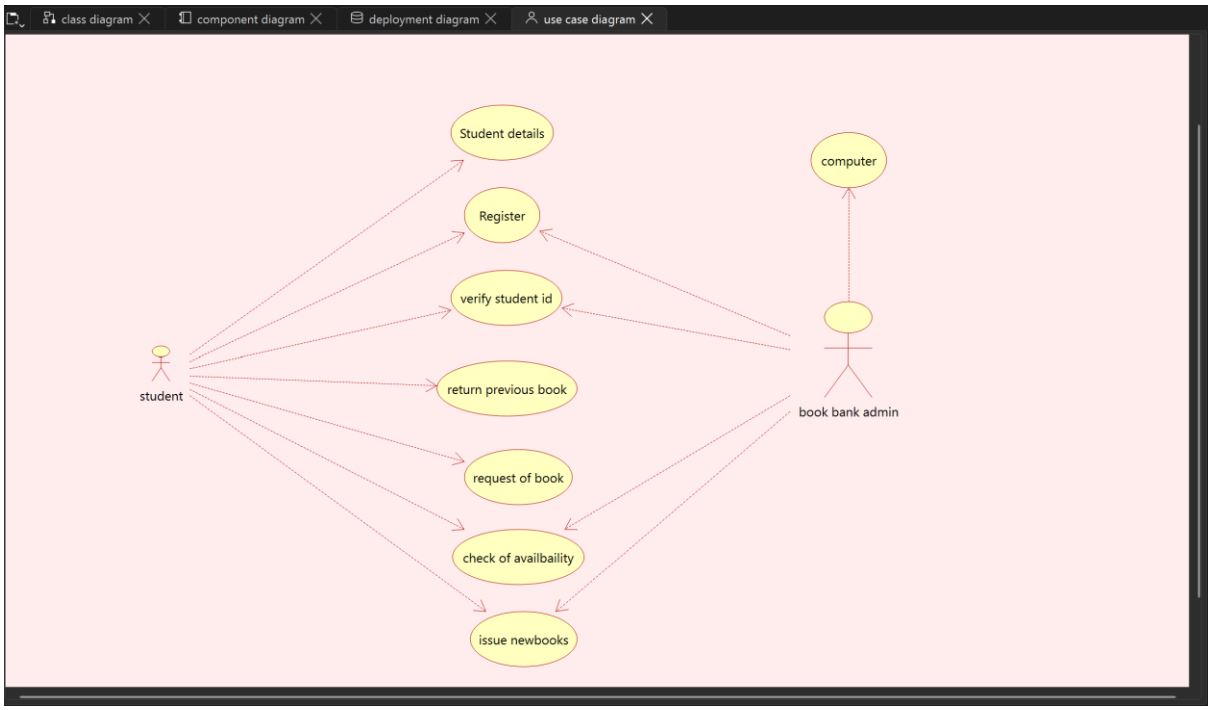
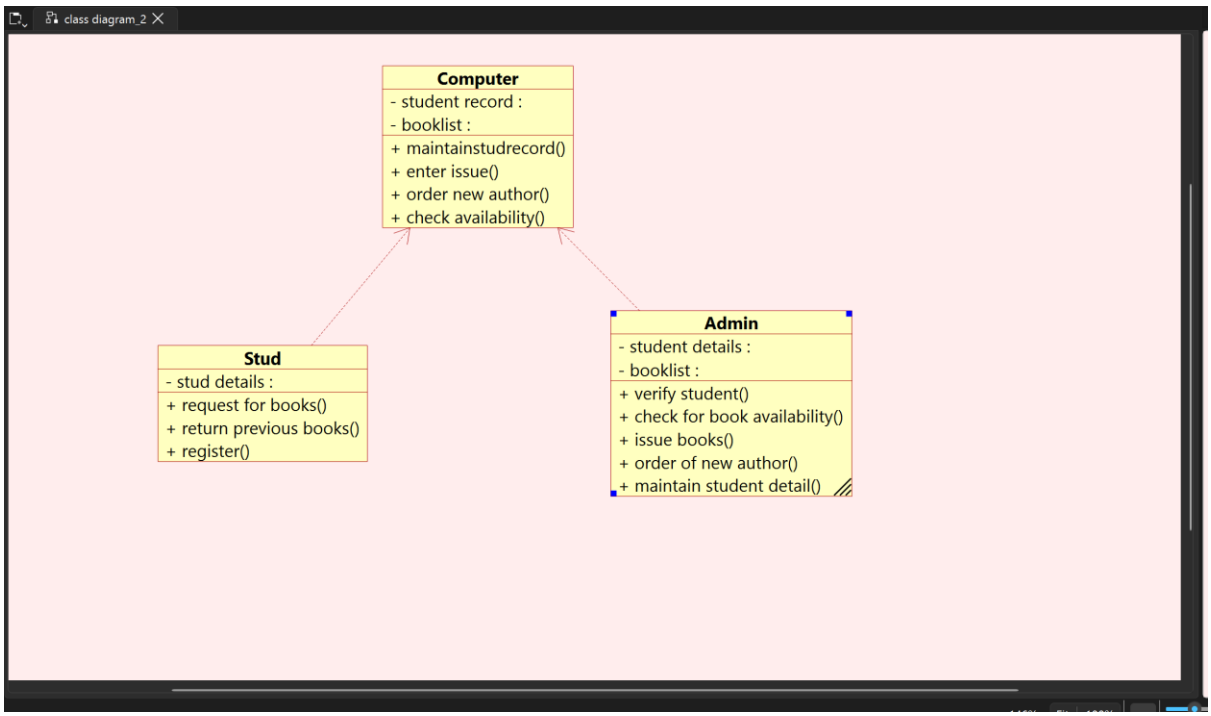
Ex No: 2 – BOOK BANK REGISTERATION SYSTEM

NAME: HEMALAKSHMI H (192571049)

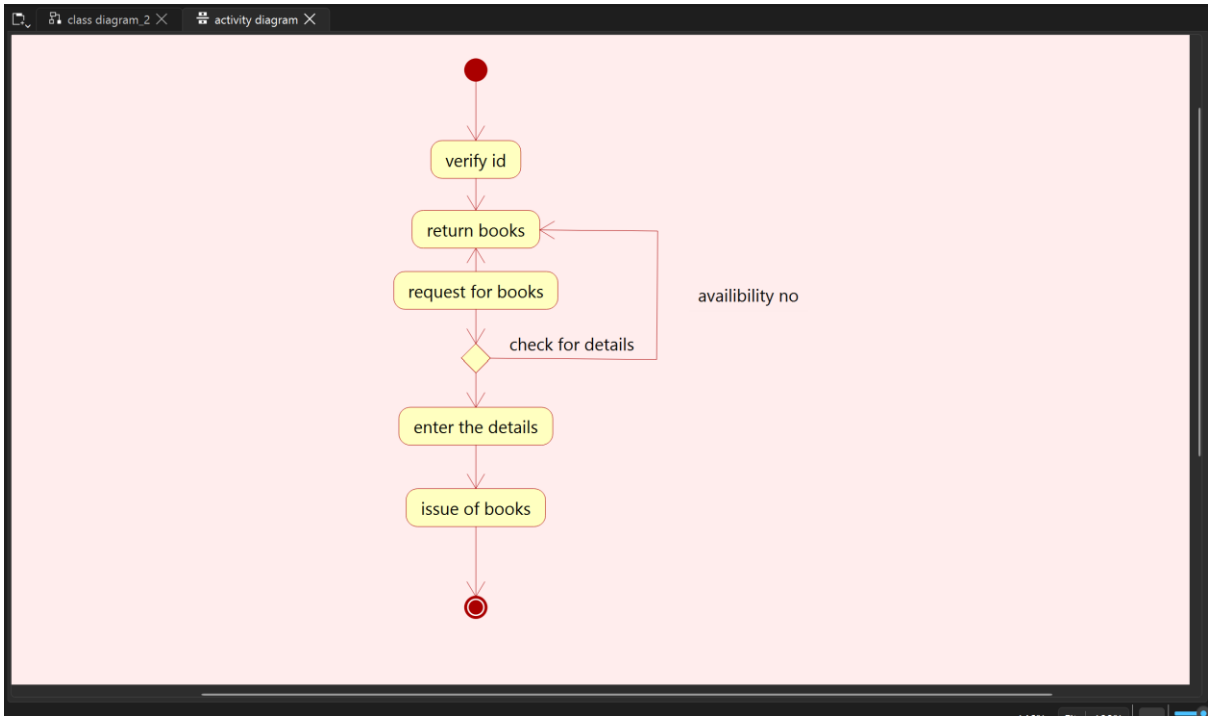
USE CASE DIAGRAM:



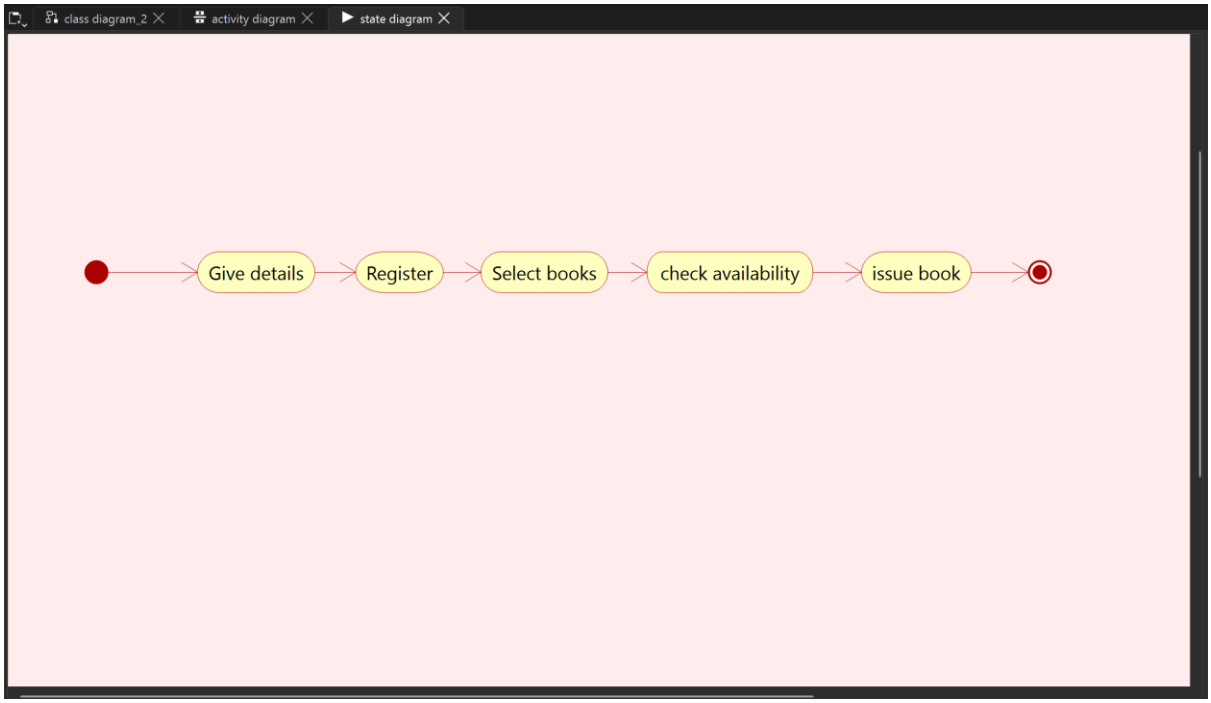
CLASS DIAGRAM:



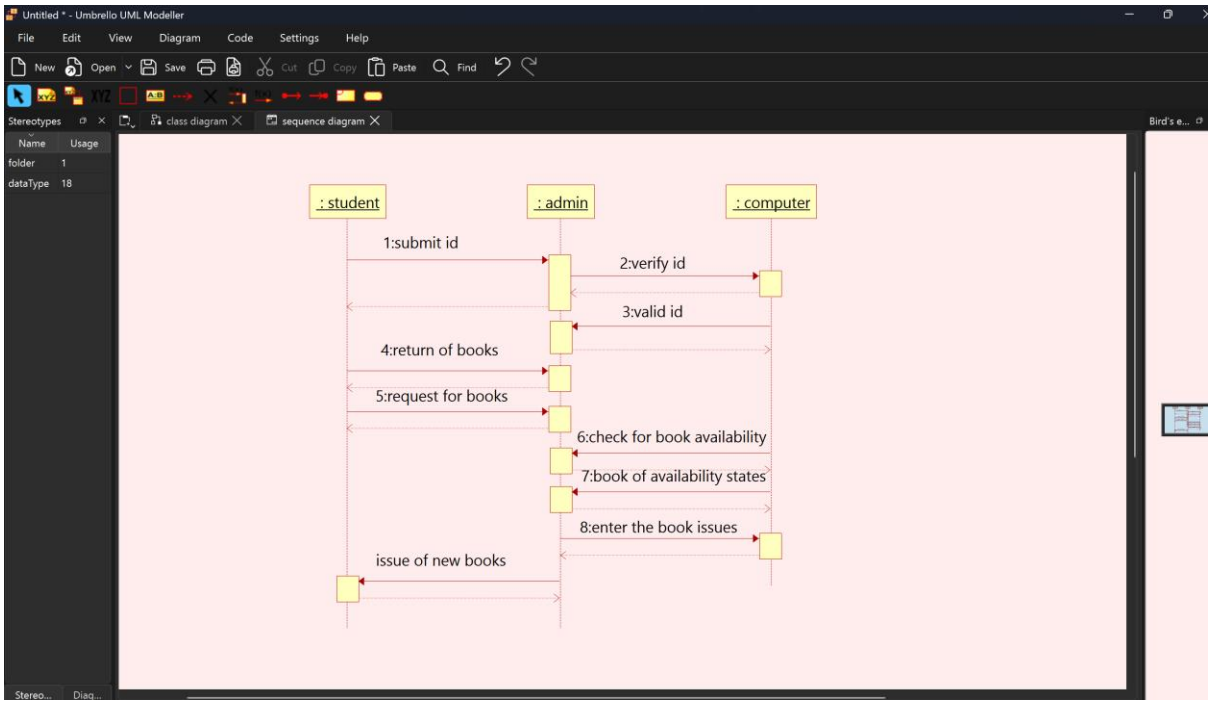
ACTIVITY DIAGRAM:



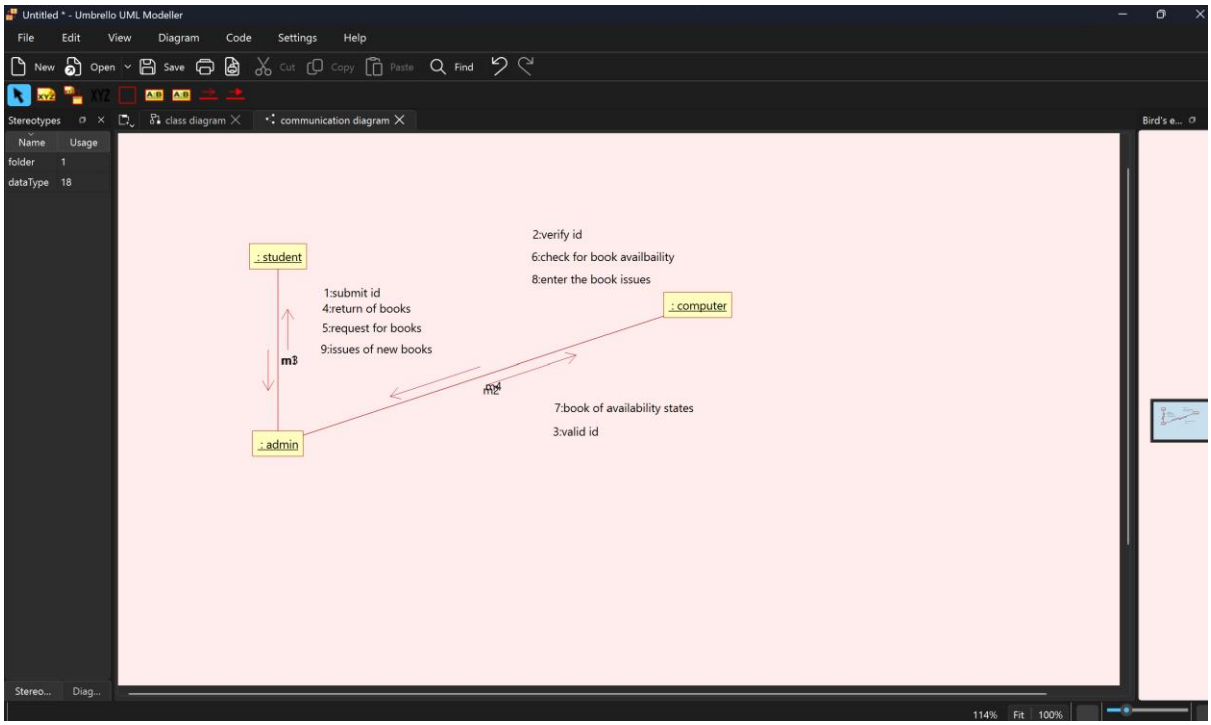
STATE DIAGRAM:



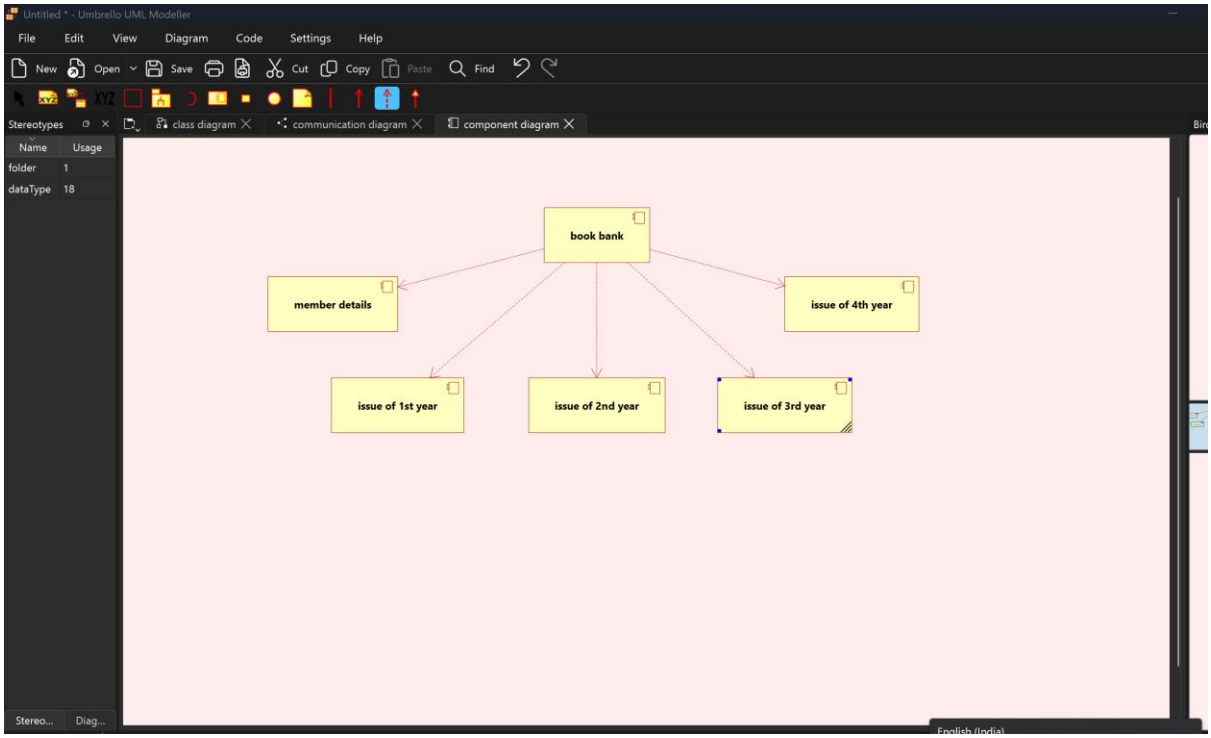
SEQUENCE DIAGRAM:



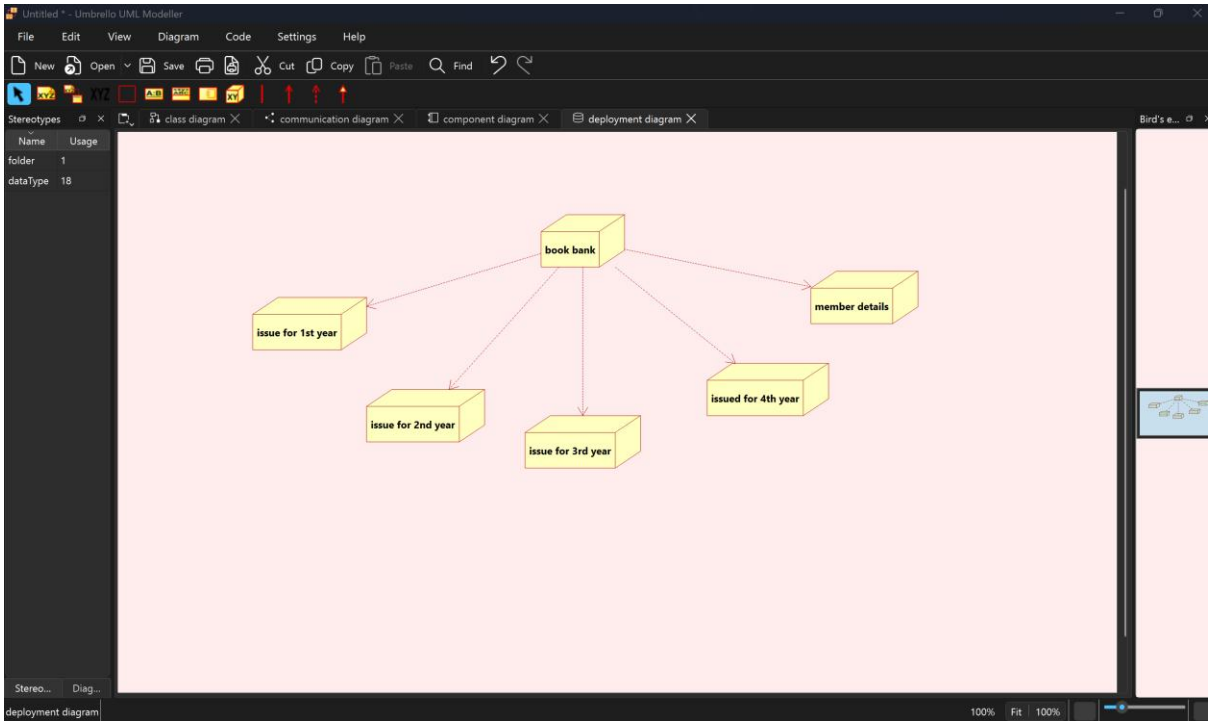
COLLABORATION DIAGRAM:



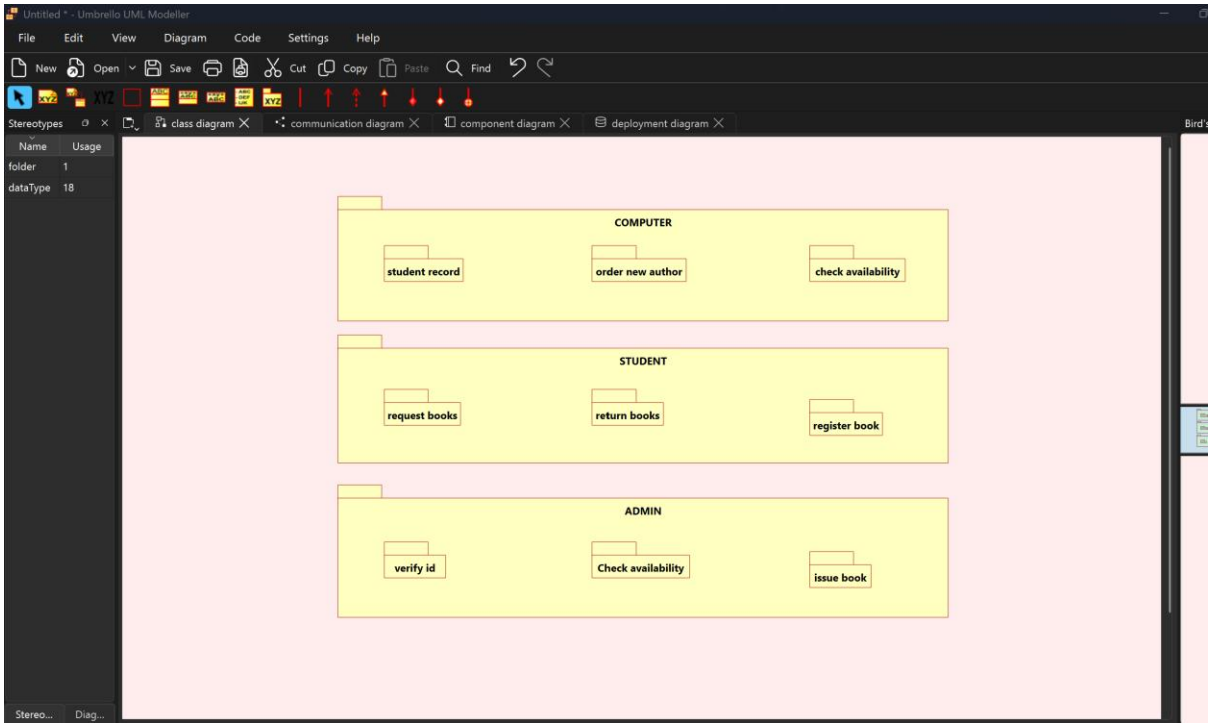
COMPONENT DIAGRAM:



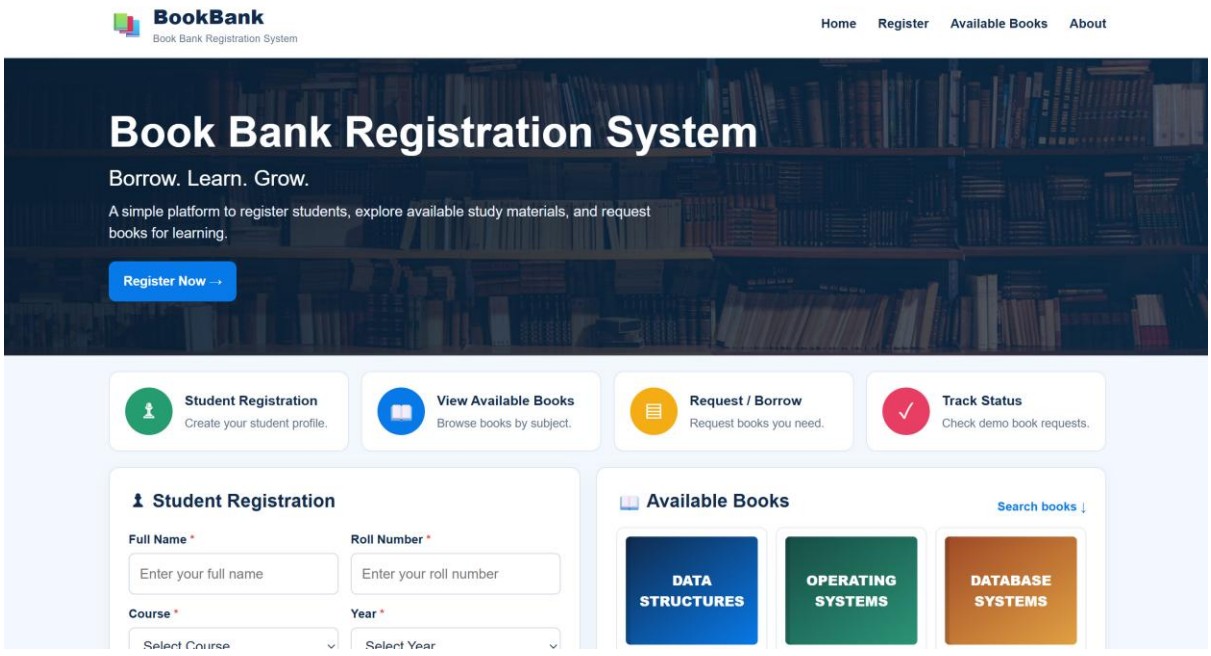
DEPLOYMENT DIAGRAM:

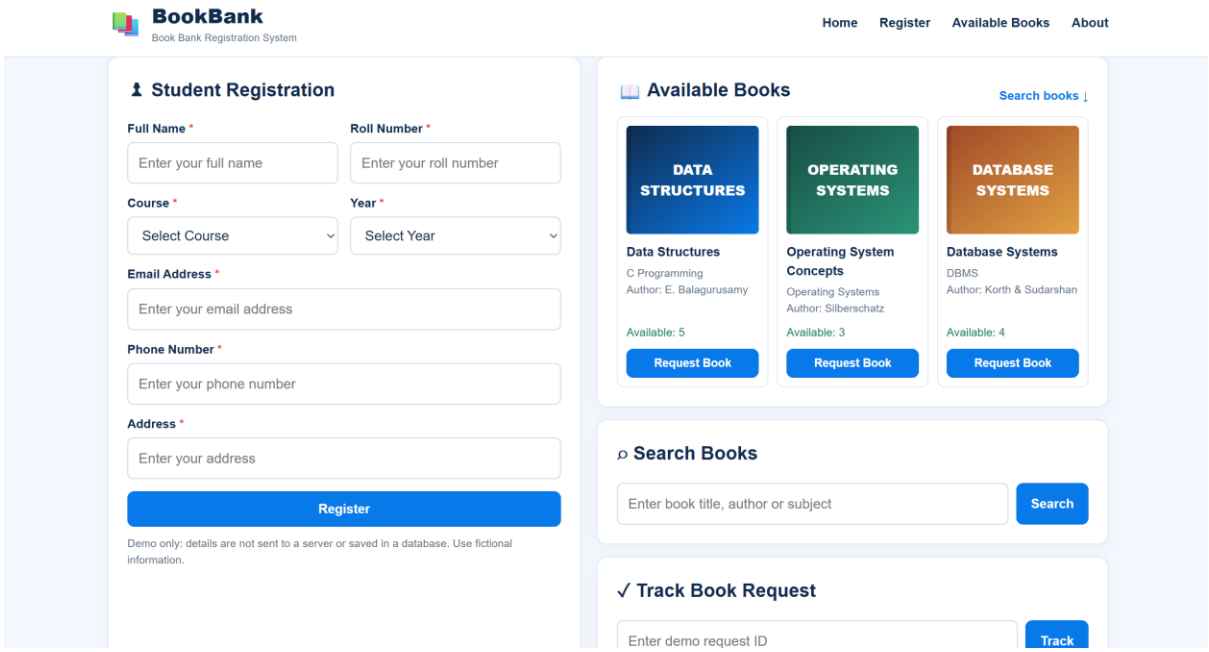


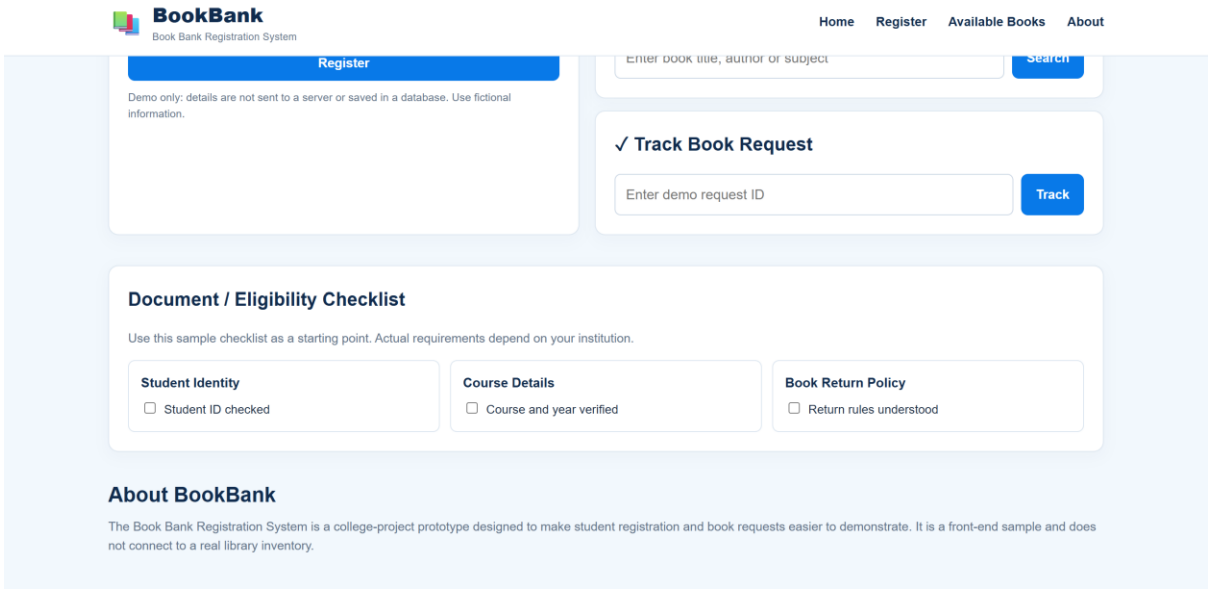
PACKAGE DIAGRAM:



WEBPAGE FOR BOOK BANK REGISTRATION SYSTEM




BookBank
Book Bank Registration System
Home
Register
Available Books
About
Student Registration
Full Name *
Enter your full name
Roll Number *
Enter your roll number
Course *
Select Course
Year *
Select Year
Email Address *
Enter your email address
Phone Number *
Enter your phone number
Address *
Enter your address
Register
Demo only: details are not sent to a server or saved in a database. Use fictional information.
Available Books
Search books ↓
DATA STRUCTURES
Data Structures
C Programming
Author: E. Balagurusamy
Available: 5
Request Book
OPERATING SYSTEMS
Operating System Concepts
Operating Systems
Author: Silberschatz
Available: 3
Request Book
DATABASE SYSTEMS
Database Systems
DBMS
Author: Korth & Sudarshan
Available: 4
Request Book
Search Books
Enter book title, author or subject
Search
Track Book Request
Enter demo request ID
Track


BookBank
Book Bank Registration System
Home
Register
Available Books
About
Register
Demo only: details are not sent to a server or saved in a database. Use fictional information.
Search
Track Book Request
Enter demo request ID
Track
Document / Eligibility Checklist
Use this sample checklist as a starting point. Actual requirements depend on your institution.
Student Identity
Student ID checked
Course Details
Course and year verified
Book Return Policy
Return rules understood
About BookBank
The Book Bank Registration System is a college-project prototype designed to make student registration and book requests easier to demonstrate. It is a front-end sample and does not connect to a real library inventory.

HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Book Bank Registration System</title>

<style>
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #f2f8fd;
  color: #203047;
}

header {
  background: #102b4e;
  color: white;
  padding: 18px 5%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
}
```

```
header h2 {
  color: #f5b544;
  margin: 0;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 8px;
}

.hero {
  background: linear-gradient(120deg, #102b4e, #0879e8);
  color: white;
  text-align: center;
  padding: 65px 20px;
}

.hero h1 {
  font-size: 36px;
}

.hero p {
  font-size: 18px;
}

button {
  background: #0879e8;
  color: white;
  padding: 12px 18px;
```

```
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-weight: bold;
}

.hero button {
  background: #f5b544;
  color: #102b4e;
}

button:hover {
  opacity: 0.85;
}

section {
  padding: 35px 5%;
}

h2 {
  color: #102b4e;
}

.container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
  gap: 20px;
}

.card, .form-box {
  background: white;
```

```
  padding: 22px;
  border-radius: 10px;
  box-shadow: 0 4px 15px #00000010;
}

.card h3 {
  color: #0879e8;
}

.form-box {
  max-width: 650px;
  margin: auto;
}

label {
  display: block;
  margin-top: 12px;
  font-weight: bold;
}

input, select {
  width: 100%;
  padding: 11px;
  margin-top: 5px;
  border: 1px solid #ccd5e0;
  border-radius: 6px;
}

.form-box button {
  width: 100%;
  margin-top: 18px;
```

```
}

.book button {
  margin-top: 10px;
  width: 100%;
}

.result {
  background: #e5f7ed;
  padding: 12px;
  border-radius: 6px;
  margin-top: 15px;
  overflow-wrap: anywhere;
}

.hidden {
  display: none;
}

footer {
  background: #102b4e;
  color: white;
  text-align: center;
  padding: 20px;
}

@media (max-width: 600px) {
  header {
    justify-content: center;
    gap: 15px;
  }
```

```
 nav {
  text-align: center;
 }

 .hero h1 {
  font-size: 28px;
 }
}
</style>
</head>

<body>

<header>
 <h2>📚 BookBank</h2>
 <nav>
  <a href="#home">Home</a>
  <a href="#register">Register</a>
  <a href="#books">Books</a>
  <a href="#track">Track</a>
 </nav>
</header>

<!-- Home Page -->
<div class="hero" id="home">
 <h1>Book Bank Registration System</h1>
 <p>Borrow. Learn. Grow.</p>
 <p>A simple platform to register students and request study books.</p>
 <button onclick="goTo('register')">Register Now</button>
</div>
```

```
<!-- Features -->
<section>
 <h2>Our Features</h2>

 <div class="container">
  <div class="card">
   <h3> 👨‍💻 Student Registration</h3>
   <p>Register student details easily.</p>
  </div>

  <div class="card">
   <h3>📖 Available Books</h3>
   <p>View books available for borrowing.</p>
  </div>

  <div class="card">
   <h3>📚 Book Request</h3>
   <p>Submit requests for required books.</p>
  </div>

  <div class="card">
   <h3>🔍 Track Status</h3>
   <p>Check the status of book requests.</p>
  </div>
 </div>
</section>

<!-- Student Registration -->
<section id="register">
```

```
<h2 style="text-align:center">Student Registration</h2>

<div class="form-box">
 <form id="registerForm">

  <label for="name">Full Name</label>
  <input id="name" required placeholder="Enter full name">

  <label for="roll">Roll Number</label>
  <input id="roll" required placeholder="Enter roll number">

  <label for="course">Course</label>
  <select id="course" required>
   <option value="">Select Course</option>
   <option>BCA</option>
   <option>B.Sc. Computer Science</option>
   <option>B.Com.</option>
   <option>B.A.</option>
   <option>B.Tech.</option>
   <option>Other</option>
  </select>

  <label for="year">Year</label>
  <select id="year" required>
   <option value="">Select Year</option>
   <option>First Year</option>
   <option>Second Year</option>
   <option>Third Year</option>
   <option>Fourth Year</option>
  </select>
```

```
    <label for="email">Email Address</label>
    <input id="email" type="email" required
        placeholder="Enter email address">

    <label for="phone">Phone Number</label>
    <input id="phone" type="tel" required
        placeholder="Enter phone number">

    <label for="address">Address</label>
    <input id="address" required placeholder="Enter address">

    <button type="submit">Register Student</button>
   </form>

   <div id="registrationResult" class="result hidden"></div>
  </div>
</section>

<!-- Available Books -->
<section id="books">
  <h2 style="text-align:center">Available Books</h2>

  <div class="container">

   <div class="card book">
    <h3> 📘 Data Structures</h3>
    <p>Author: E. Balagurusamy</p>
    <p>Subject: Computer Science</p>
    <p>Available Copies: <span id="qty1">5</span></p>
    <button onclick="requestBook('Data Structures', 'qty1')">
     Request Book
```

```
    </button>
   </div>

   <div class="card book">
    <h3> 📗 Operating Systems</h3>
    <p>Author: Silberschatz</p>
    <p>Subject: Computer Science</p>
    <p>Available Copies: <span id="qty2">3</span></p>
    <button onclick="requestBook('Operating Systems', 'qty2')">
     Request Book
    </button>
   </div>

   <div class="card book">
    <h3> 📕 Database Systems</h3>
    <p>Author: Korth</p>
    <p>Subject: Database Management</p>
    <p>Available Copies: <span id="qty3">4</span></p>
    <button onclick="requestBook('Database Systems', 'qty3')">
     Request Book
    </button>
   </div>

  </div>

  <div id="bookResult" class="result hidden"></div>
</section>

<!-- Search Books -->
<section id="search">
  <h2>Search Books</h2>
```

```
 <div class="form-box">
  <input id="searchInput"
       placeholder="Enter book title or subject">

  <button onclick="searchBooks()" style="margin-top:12px">
   Search
  </button>

  <div id="searchResult" class="result hidden"></div>
 </div>
</section>

<!-- Track Requests -->
<section id="track">
 <h2 style="text-align:center">Track Book Request</h2>

 <div class="form-box">
  <form id="trackForm">
   <label for="requestId">Request ID</label>
   <input id="requestId" required
        placeholder="Enter your request ID">

   <button type="submit">Check Status</button>
  </form>

  <div id="trackResult" class="result hidden"></div>
 </div>
</section>

<footer>
```

```
  <p>© 2026 Book Bank Registration System</p>
  <p>College Project | Borrow • Learn • Grow</p>
</footer>

<script>
let studentRegistered = false;
let studentRoll = "";
let requests = [];
let nextId = 1001;

function goTo(id) {
  document.getElementById(id).scrollIntoView({
    behavior: "smooth"
  });
}

// Student Registration
document.getElementById("registerForm")
.addEventListener("submit", function(event) {
  event.preventDefault();

  const name = document.getElementById("name").value.trim();
  studentRoll = document.getElementById("roll").value.trim();

  studentRegistered = true;

  const result = document.getElementById("registrationResult");
  result.classList.remove("hidden");
  result.textContent =
    "Registration successful! Welcome, " + name +
    ". You can now request books.";
```

```
});

// Request a Book
function requestBook(bookName, quantityId) {
 if (!studentRegistered) {
  alert("Please register as a student first.");
  goTo("register");
  return;
 }

 const quantityElement = document.getElementById(quantityId);
 let quantity = Number(quantityElement.textContent);

 if (quantity <= 0) {
  alert("This book is currently unavailable.");
  return;
 }

 const id = "BB-REQ-" + nextId++;

 requests.push({
  id: id,
  book: bookName,
  roll: studentRoll,
  status: "Request Submitted"
 });

 quantityElement.textContent = quantity - 1;

 const result = document.getElementById("bookResult");
 result.classList.remove("hidden");
```

```
  result.textContent =
    "Book request submitted for " + bookName +
    ". Your Request ID is " + id;
}

// Search Books
function searchBooks() {
  const query = document.getElementById("searchInput")
    .value.trim().toLowerCase();

  const books = document.querySelectorAll(".book");
  let found = [];

  books.forEach(function(book) {
    const matches = book.textContent.toLowerCase().includes(query);
    book.style.display = matches ? "" : "none";

    if (matches) {
      found.push(book.querySelector("h3").textContent);
    }
  });

  const result = document.getElementById("searchResult");
  result.classList.remove("hidden");
  result.textContent = found.length
    ? "Books found: " + found.join(", ")
    : "No matching books found.";
}

// Track Book Requests
document.getElementById("trackForm")
```

```
.addEventListener("submit", function(event) {
 event.preventDefault();

 const id = document.getElementById("requestId")
  .value.trim().toUpperCase();

 const request = requests.find(function(item) {
  return item.id.toUpperCase() === id;
 });

 const result = document.getElementById("trackResult");
 result.classList.remove("hidden");

 if (request) {
  result.textContent =
   "Request ID: " + request.id +
   " | Book: " + request.book +
   " | Status: " + request.status;
 } else {
  result.textContent =
   "Request not found. Submit a book request first.";
 }
});
</script>

</body>
</html>
```